1. Introduction

Definition 1.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 1.2 (Central factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)\left(n + \frac{k}{2} - 2\right)\cdots\left(n - \frac{k}{2} + 1\right) = n\prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right), & \text{if } k > 0. \end{cases}$$

Proposition 1.3 (Central Newton's formula)**.** *For arbitrary integers x, and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!}\delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference of $f(t)$, defined by

$$\delta^k f(t) = \sum_{j=0}^{k}(-1)^j \binom{k}{j} f(t + \tfrac{k}{2} - j).$$

Proof. See [**?**, p. 462]. □

Lemma 1.4 (Central falling factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0. \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right)$ are falling factorials.

Lemma 1.5 (Binomial form of central factorials)**.** *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n + \frac{k}{2} - 1}{k - 1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!}\left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!}\left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$, and Lemma (1.4). □

Lemma 1.6 (Halved central factorials)**.** *For integers n, k,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\binom{n+\frac{k}{2}}{k} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \binom{n+\frac{k}{2}}{k} - \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\left[2\binom{n+\frac{k}{2}}{k} - \binom{n+\frac{k}{2}-1}{k-1}\right] = \frac{1}{2}\left[\binom{n+\frac{k}{2}}{k} + \binom{n+\frac{k}{2}-1}{k} + \binom{n+\frac{k}{2}-1}{k-1} - \binom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Proposition 1.7 (Central Newton's formula for powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!}\delta^k t^m.$$

Proposition 1.8 (Generalized hockey-stick identity)**.** *For integers a, b and j,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j}$, which means that, $\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$